‘LIVING IN A PERFORMING WORLD – PERFORMING IN A LIVING WORLD’

Simon Emmerson

Music, Technology and Innovation Research Centre
De Montfort University, Leicester, UK

ABSTRACT

There has been a steady erosion of the ‘living/non-living’ dichotomy (of which ‘organic/inorganic’ is a version). This has given way to a more ecological model in which ‘living’ becomes an activity rather than a state [Ingold - 6]. Until a century or so ago the activity was limited to the cause/effect of human body action, but the sounding effect may now be displaced entirely from the original cause, both in time and space. This is ‘action over a distance’ rather than ‘dislocation’ – and technology might become the healer. Due to recording, the sounding world itself has become an ever wider source of musical material. More recently it has made a transition to becoming a fully fledged ‘musical instrument’. I will suggest that in this mediatised world ‘liveness’ is a network of activity in which subjects (performers) and objects (performed), causes and effects, can never unambiguously be ‘sorted out’ (our musical equivalent of the Turing test may not be very conclusive). Ingold’s point is that in many societies apparently inorganic (‘non-living’) objects *are* imbued with a spirit – an animistic world – which may be related to in the same way as to fellow creatures. Clearly the Schaefferian world of *reduced listening* has here transformed. How do we feel when we activate this world *as live music*? All kinds of causes may be evoked - breath, friction, percussion are inherited from the mechanical world, to which bio-activity and perhaps imagination itself may be harnessed to play our ‘animate net-work’.

1. EVOLUTION, PROGRESS AND ODYSSEY

This paper aims to be evolutionary. There is a clear narrative of journey – even an Odyssey (appropriate for my attendance at this conference). But I am not so sure there is in the end any goal – that’s one distinction between progress and evolution. I am seeking to understand a bit better where we are and what problems and opportunities we face. As the world evolves so must our ideas. So the first part of my talk brings together ideas I have thought and written about over many years but (I hope) in a way which sets the stage for another nudge along the way.

My own attempts to pin down the ‘live’ in ‘live electroacoustic music’ have changed over the years. When I was a student (in the late 1960s to early 1970s) it meant a live performer with a traditional acoustic instrument or voice, the sounds transformed in some (analogue) way and projected on loudspeakers. The French coined the term ‘mixed music’ for the combination of such a performer (transformed or simply amplified) with pre-recorded sound then on tape. In the 1980s I hated and resisted the slow and steady invasion of the term ‘real time’ which came from the earliest event-based (Midi) computer processes – I wrote about this as a binary opposition in the article ‘Live versus Real-Time’ [3].

As computers became powerful enough to combine event and signal processing, mixing and diffusion (as well as becoming more portable) in the 1990s, I stubbornly retained my views on the nature of ‘live’ against my more adventurous students and other practitioner colleagues – even when the term ‘live’ regained some standing once again in the last years of the century. DJs and laptop improvisers were called ‘live’ which was a step forward in the rehabilitation of the word – but inaccurately applied I thought. With a lot of what they were doing (no conventional instrument or voice in sight) I remarked ‘that’s not live!’ to which they steadfastly replied ‘yes it is!’. So to celebrate the new century I reconsidered my position and wrote in *Living Electronic Music* [4: 90] -

Working Definition of ‘Live’ (for the Time Being)

We need a working definition, not to define boundaries for all time, but to set the limits of the stage we will occupy for the time being.

‘Live’ will here mean:

- The presence of a human performer:

who takes decisions and/or makes actions during a performance which change the real sounding nature of the music;

- this embraces the historically accepted view of the ‘live’ as involving a human:

who produces sound mechanically; or who produces sounds on electronic substitutes for mechanical instruments using similar physical gestural input;

- but it also includes one:

  who does not mechanically cause the sound, yet who may cause, form or influence it through electronically mediated interfaces under their immediate control.

Today I want to shift territory once again – from observer to participant, from musicologist to performer. This temporary working definition just cited still focused on the world of the performer on stage, being listened to and observed by me – by us. This time around we live in a world as contradictory as ever - one more connected up, yet more distributed, in touch when far apart, yet still wanting human warmth and presence, creating 'virtual realities' while still seeking the noises and smells of the rain forest and the feel of warm wind.

I welcome the combination of 'organic' and 'live' in the theme of the conference. There has been a steady erosion of the 'living/non-living' dichotomy (of which 'organic/inorganic' is a version) which mirrors the later 20thC rejection of 'mind/body', 'natural/cultural' and other binary thinking. This has given way to a more ecological (holistic) model which seeks to place *interdependencies* more clearly to the fore.

> For life is not a principle that is separately installed inside individual organisms, and sets them in motion upon the stage of the inanimate. To the contrary […] life is 'a name for what is going on in the generative field within which organic forms are located and 'held in place' [6: 200]

In this (Ingold's) view 'living' becomes an activity rather than a state. Music is a very special subset of this activity – one which seems to have 'added meaning' as well as 'added value'. Furthermore technology has extended its range and ubiquity. Until a century or so ago the activity was limited to the cause/effect of human body action, but the sounding effect may now be displaced entirely from the original cause, both in time and space. This is a kind of musical or sound prosthesis allowed by technology and has progressively become global.

2. DISLOCATION AS *ENTFERNUNG* NOT *VERRENKUNG*!

Let us go back to what I originally wrote. In 1994 I wanted to look again at the Schaefferian idea of the 'acousmatic'. I was a bit unhappy with the simple Pythagorean idea of a 'venerated lecturer behind a curtain'. In fact Pierre Schaeffer's discussion seemed to revolve around a group of consequences of the development of 'technologies of sound'. I called these the 'acousmatic dislocations':

> Dislocation 1: Time (recording)
> Dislocation 2: Space (telecommunications (telephone, radio), recording)
> Dislocation 3: Mechanical causality (electronic synthesis, telecommunications, recording) [4: 95]

We can rewrite these in terms of the extraordinary final quarter of the 19th century -

> Telephone: Alexander Graham Bell (1876) - separating source and receiver in space.
> Sound recording: Thomas Edison (1877) - separating source and receiver in time and space.
> Radio: Marconi (1890s) - radio waves - across the Atlantic (1901).
> Synthesis: Elisha Gray – Thaddeus Carhill (1870s-90s) – separating sound production (as perceived) from immediate mechanical causality[1].

In 1996 I was invited to the *Digitale* conference and festival in Cologne. I speak basic German and risked presenting my paper *auf Deutsch* – I spent quite some time with dictionary in the preparation and enjoyed the experience. In fact it was a simple development of the 1994 paper, so of course contained the phrase 'acousmatic dislocation', and I translated this as *akusmatische Verrenkung*. Now *Verrenkung* is fine for dislocation if you mean you need a doctor to get your shoulder back into place, and I was met with at best humour and at worst incomprehension in using this phrase. I was told that speaking figuratively the German language would switch to something like the much more poetic and abstract *Entfernen* or *Entfernung* – which I would retranslate as *removal* or *distancing*. Now this led me while thinking about this paper (here) to an interesting insight – that we can make 'lost in translation' into something gained by correction and comparison.

The 19th century saw the creation of a EuroAmerican-centric order based in large part upon that very idea – a *distancing* not a *dislocation* - not so much 'something that needs correcting' as something that need traversing. The technological discoveries just referred to can be re-read as not so much the *causes* of any dislocation but quite the opposite – an answer to them. Technology reaching out across these gaps – oceans of space and time that our imagination had longed for centuries could be crossed.[2] We always did travel to far off lands and so the *dislocations* were always there - but maybe we wanted to see and hear those we had left behind. Such technology has the potential to bridge the gap – to heal the rift – to link together. Thus my list above may be turned on its head and seen as the start of a *creative response* to dislocation – not its *cause*. Sending letters was a great start after the invention of the alphabet – but to preserve and hear the absent voice (in both distance and time) was a dream that Edison specifically addressed at the time of his invention.

[1] That is we perceive (say) what we believe to be a human voice from the vibrations of a loudspeaker cone which remains mechanical.

[2] See the oft quoted Francis Bacon, writing in 'New Atlantis' (1624/27) "We have sound houses, where we practise and demonstrate all sounds and their generation. … We have all means to convey sounds in trunks and pipes, in strange lines and distances."

3. LISTENING, IMITATING, RECORDING AND CAPTURE

In the course of the twentieth century we have seen a steady transformation from the world (that is our environment) as the source of musical sound to the world itself as potential performance instrument. For centuries (millennia we might suppose) music has imitated the sounds around us – usually the sounds of living creatures but not always. But through recording, the sounding world itself has become an ever wider source of musical material well beyond what we can mechanically imitate with our voices or objects around us, projected into a variety of acoustics. The early 20th century futurists ensured that we consider all sounds including the most noisy and industrial – but before them one wonderful precursor:

> At a sufficient distance over the woods this sound [bells] acquires a certain vibratory hum, as if the pine needles in the horizon were the strings of a harp which it swept. All sound heard at the greatest possible distance produces one and the same effect, a vibration of the universal lyre. (Thoreau) [13: 168]
>
> To convince ourselves of the amazing variety of noises, it is enough to think of the rumble of thunder, the whistle of the wind, the roar of the waterfall, the gurgling of a brook, the rustling of leaves, the clatter of a trotting horse [...], and of the generous, solemn white breathing of a nocturnal city. (Russolo) [10: 85]
>
> […] We will sing of the vibrant nightly fervour of arsenals and shipyards blazing with violent electric moons; […] deep-chested locomotives whose wheels paw the tracks like the hooves of enormous steel horses bridled by tubing; and the sleek flight of planes whose propellers chatter in the wind like banners and seem to cheer like an enthusiastic crowd. (Marinetti) [7: 22]

Thoreau's ideas fed through of course to Cage and beyond - while the futurists' *intonarumori* had more bark in the telling than bite in their realization (judging by the few extant recordings and recreations). Then there was Arseny Avraamov's *Symphony of sirens* most famously performed in the city of Baku in 1922, to celebrate the fifth anniversary of the Russian October Revolution in 1917 (also performed a year later in Moscow) which has recently been thoroughly documented and recreated by a group of Spanish artists directed by Miguel Molina Alarcón [8][3]:

> […] Avraamov worked with choirs thousands strong, foghorns from the entire Caspian flotilla, two artillery batteries, several full infantry regiments, hydro-airplanes, twenty five steam locomotives and whistles and all the factory sirens in the city. He also invented a number of portable devices, which he called "Steam Whistle Machines" for this event, consisting of an ensemble of 20 to 25 sirens tuned to the notes of the *Internationale*. […] Avraamov did not want spectators, but intended the active participation of everybody […][4]. ([8: 19]

Avraamov himself wrote –

> The social problems of music can be solved in one of two ways: the active engagement of the broad masses with musical creativity (the socialization of music) or the modernization of its hitherto archaic material technology (the electrification of music). [He goes on to refer to the dream of] the creation of new, powerful, instruments […] that could sound simultaneously for millions of listeners. (Arseny Avraamov (orig. 1924 translated in [8: 70])

In 1986 I wrote about *mimesis* in electroacoustic music contrasting *timbral mimesis* in which the sound quality of something was imitated with *syntactic mimesis* in which a *pattern of relationships*[5] was modeled in sonic terms [2]. Of course the two are combined in raw environmental phenomena – but to a surprising extent they can be separated – as in the famous experiments on 'bouncing' and 'smashing' behaviours [15].

A very large part of algorithmic composition might be seen as a kind of *behaviour mimesis*. From environmental measurements of cosmic events to weather maps, through animal and human behaviours, from deepest rural to most complex urban phenomena – anything that may be described in mathematical symbolic form can 'drive' a synthesis environment. Iannis Xenakis is often cited as one of the first to harness the power of the *world's behaviours* to these ends - in Xenakis's case not (literally anyway) the *world's sounds*.

Thus a divided world of mimesis emerged in the last half of the 20th century. Behaviours and sounds became strangely separated when abstracted from the world's near infinite resources now available through recording and analysis. In many ways this exemplified the perennial argument, recurring over many generations but clearly witnessed in the contemporary music battles of the 1950s and 1960s, between those who composed drawing out relationships primarily from sound quality ('abstracted syntax') and those who relied to an extent on *a priori* symbolic/numerical 'systems' ('abstract syntax') [2]. The promise that empirical research might come to the rescue of this split way of thinking has not been delivered to any great extent. In this view research in psychology of perception (including sound behaviour)

[3] Molina Alarcón's book (with CDs) covers a lot more than this – it is subtitled 'Original documents and reconstructions of 72 key works of music, poetry and agitprop from the Russian Avantgardes (1908-1942)'.

[4] These are Molina's words but followed by the complete instructions for the performance written by Avraamov himself [8: 20-21).

[5] The word *behaviour* can be used here – although I did not use it when originally conceived. It is also the term used by Denis Smalley for one of his nine indicative fields [12].

could help us devise 'rules' – not idealized ones but strictly empirical, derived from observed examples – which might drive artificial intelligence applications finding 'suitable' sounds – that is sounds that 'suit' a particular behaviour.[6] Much installation art seems to focus on such syntactic (generative) concerns at the expense of the sound quality.

What I am suggesting here is the possibility of bringing two sometimes distinct areas of sonic endeavour together – the installation and the performance. I will suggest that in this mediatised world 'liveness' is a network of activity in which subjects (performers) and objects (performed), causes and effects, can never unambiguously be 'sorted out'. Liveness becomes a property of relationship and response, not something which resides solely in decoding 'the sound itself'.

4. LIVE – A LISTENER'S OR A PERFORMER'S PERSPECTIVE

In a high proportion of what I have written on liveness in electroacoustic music I have focused on 'decoding the sound stream' – that is trying to set the questions on 'living presence' almost like a guessing game – what are the cues and clues we have available to make an intelligent assessment of the agent at work producing the musical sound [4]. As I discussed above this tends to reflect an observer's perspective rather than that of an active and influential participant.

Thus in a more recent chapter [5] I looked again at liveness from the specific point of view of performance and 'bodily expression'. For this I revisited Alan Turing's famous test for machine intelligence (he did not use the term 'artificial'). This has been much simplified in almost every description of it[7]. I will not re-run that argument here except to say that it involves a complex game of playacting. It is also strongly *behaviourist* – that is if the machine behaves in a certain way it might be described as 'intelligent' – that is not to say it has reached decisions in the same way as you or I!

There is no direct and easy mapping of this to music performance – but some of the principles hold good. Let us imagine an internet mediated jam session. Dislocated in our old sense but now re-located by technology.

'H' is our real human and 'H?' may be human but then again may be a machine. In the first version 'I' (originally 'I for Interrogator' but it can be I myself), am observing the session (but cannot see the 'H'/'H?' participants through 'the curtain') and can begin to ask questions –

- firstly and most importantly is the music any good? Are the participants responding appropriately in the musical context?

And then I might risk the more subtle question –

- are the participants human or machine?

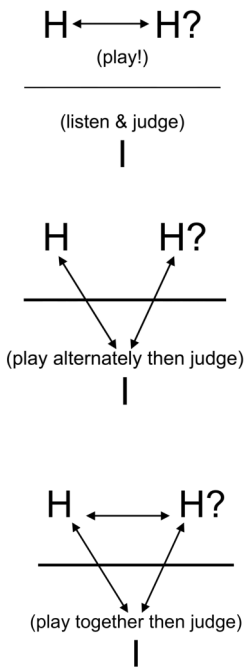


Figure 1: Rethinking Turing's Game for Music (three versions)

In the second set-up I join in, playing with each of the other participants in turn (I am clearly myself a participant now) and then judge likewise, and then finally as a trio (version 3) and ask the same questions. But in versions 2 I have not just observed 'response between others' but felt it as a 'response to me', combining the two questions in version 3.

This is a fundamental shift – one I was not so aware of when I wrote a first version of this paper in 2005 for the Live Algorithms for Music Conference in London. We will all be familiar with the recent ideas that listeners are clearly participants in music making. And ethnomusicologists have been aware for many decades that there is no such thing as an observer who is totally independent of (and so does not influence) the observed. That said we have a difference of degree which is important for musicians – a participant who actively seeks to create or to manipulate sounds to contribute to a performance is clearly involved in a more direct way than a participant listener.

When we join in *fully as performers* we shift the centre of gravity in an ensemble – we give and (usually) expect to receive – that is we expect a *response on a personal level*. We have clearly moved away from judging the sound as an observer – 'is that live?' – to judging as a participant performer – 'am I getting a response?'.

In Emmerson 2007 [4: 3] I divided the world of 'living presence' in sound into three 'worlds' – the first

[6] There is some evidence that such sophisticated research is being applied steadily in the computer games industry [1].

[7] It is much more than testing to see if a machine behaves 'indistinguishably' from a human – the experiment asks the machine to deceive us and we measure if it has succeeded as well as or better than a 'real' human! [14].

physical and mechanical (perceiving action and agency), the second psychological (working out ideas of will, choice and intention); both these enwrapped by a third 'personal and social presence (what you mean to me, where I am and who I am with)'[8]. Here too, my model tended to be one of observation rather than participation. 'I am at a concert – do I believe what they are performing on stage is 'live'?'.

Let us now extend it to include ideas of *response* just examined. Much more along the lines 'I am participating – how am I relating to others and how are they relating to me?'. This is an added dimension to the previous observations and does not displace them in any way. The first world becomes that of haptics (and all other physical input to our senses) – in a world with gravity and atmosphere there is no movement without some form of resistance. The second world moves us from observing a game or reading a narrative to playing the game and constructing the narrative. At a stroke our musicology has entered the realm of the computer game like environment in which the user is clearly a performer – whose will, choice and intention directly influence the outcome.

The third world which wraps these two immersively is the personal and the social. In my original article I bracketed these two words together – but now I want to separate them again. Personal – that is individual – sound and music space has emerged and become almost dominant since the launch by Sony of the Walkman in 1979 through to the iPod of recent years. Many computer games retain a solo emphasis – bettering your previous score rather than a simultaneous opponent – a trend broken notably by a small number of multi-user interfaces such as the Wii[9].

The *social* – community – aspects of media music making remain problematic in several senses. There are laptop orchestras. These face the problem that the laptop *is* already an orchestra – we need a kind of niche hypothesis[10] here such that each animal in the menagerie does not simply drown the others out, or ignore them completely. Then in internet collaborative music making in general, how do I gain a sense of 'others' performing with me. We might start with a kind of extended 'skype' – but this is totally inadequate in its present form. As a performer, visual and aural cues from other performers are tightly synchronized – it is only our analysis which has taken the two apart. And the intangible ability of some performers to anticipate a downbeat with apparently little by way of visual cue is well known – but only if they are in the same space together[11].

[8] Taking from Christopher Small the idea that *relationships* create the 'meaning' of musical performance [11: 93].

[9] With a maximum of 16 Wii remotes to a console.

[10] Evolution has tended to encourage a sharing of the available bandwidth in noisy environments (see the writings of Bernard L. Krause).

[11] This is not to say that tight synchronicity is a necessity for live music making – only that there will be consequences in its loss – as no network will have latency less than that determined by the speed of light.

But high quality versions of current media may function when our task is to link 'real humans' together in internet mediated traditional performance (whether on traditional instruments or electronic substitutes). But I wish to extend the idea of performing 'agent' for which this remains problematic.

5. THE ANIMATE NET-WORK

Many years ago I used to refer to one function of live electronic music as 'humanising the technology'. This shifted to thinking of it more as 'humanising through technology' – now I have arrived at describing this process as 'reanimation'. This is a small but important shift, embracing a much wider view of 'living' in the sense of activity as suggested by Ingold and in the terms discussed above.

Yet don't we seek out other 'organic' activity? – it is after all maybe more likely to respond to us. Take care here! Ingold's point (and that of many other anthropologists) is that in many societies apparently inorganic ('non-living') objects *are* imbued with a spirit – an animistic world – which may be related to in a similar way as to fellow creatures. It is this relationship activity which is 'living'. Clearly the Schaefferian world of *reduced listening* has here transformed – sound is certainly not only 'for sounds sake' in this case - it indicates far more than a cause when it indicates an object's spirit.

The final part of this paper tries to tie together the nudges forward of the several ideas I have presented over the years into one object. I shall call this the 'animate net-work'. The 'animate net-work' is a *Gedankexperiment* – that is it's a kind of imaginary beast which might exist and whose possible existence might get us exploring alternative possibilities – a *thought experiment* with (I hope) practical consequences.

Figure 2: The 'animate net-work' - an interconnecting web of agencies: human, environmental, computer-generated – this is impossible to draw but somehow we can visualize it.

Figure 2 is intended to combine some simple agencies in an imaginary group entity – it's not simply a 'work', an 'ensemble' or a 'performance' – but a kind of 'growth' which might combine elements of all three. Technology has given us the opportunity to reinstate 'animacy' to a range of agencies and activities on the network. A performance element used to be 'organic' but it is now 'animate'; everything has the potential to be thought of as animate – it need not be confined to carbon-based life forms. In this model liveness becomes responsiveness – in all the senses so far discussed. I stress – we are not discussing whether any *individual* element (agency) is alive or not. We can add the idea of *growth*, the way that nodes, that is active agents, on the network may be added or substracted (I really should use the terms 'are born' or 'die'!) – and then we have a fully fledged (strange turn of phrase!) ecosystem.

How do we (composers and performers) feel when we activate this world *as music*? The world sound bank is 'touched' and responds – *how* it responds is up to us, or another agent in the net-work.

6. PERFORMING THE ANIMATE NET-WORK

We have seen an increasingly sophisticated restoration of body senses in recent interface development – all the kinds of causes may be evoked and remodeled to give us back our sense of corporeal involvement in sound production and transformation. We can return to some fundamentals: from our first world (physical presence) we have the primary methods by which humans have provoked the environment to sound - these involve the use of muscles and the transfer of energy in the mechanical and physical domain – breath, friction and percussion.

But wait – this has until now applied to individuals. But our net-work now 'relocates' this individualism within the living world as a whole from whence it came and to which it shall return. The world is now a potential performer. So of course there are environmental examples everywhere we look and hear of these three basic energy transfers – a rainforest is jammed packed full of wind, stridulation, creaks and groans, gravity driven percussion – from animal, vegetable and mineral origins. Although there is one sound provocation which is not commonly found in traditional music production – the 'breaking' of substances in a way which leaves them changed – the crack of a branch, the sound of an earthquake, the pop of a balloon, explosions (elemental and human created - the sounds of fireworks etc.) – of course these have become common sources in electroacoustic composition but can now be restored to our repertoire of performance techniques[12].

Then from our second world (psychological presence) we can harness for our performance bio-activity – which in this context means 'the body electric', that is without perceptible muscle movement. This has been with us since the 1960s [9] but clearly steps are now being made at more sophisticated and deeper levels of neural activity – the foothills of harnessing imagination itself[13].

How might this be extended in my flexible network model – beyond the individually human? Clearly group mental activity has been the subject of much interest over recent decades (whether Jungian in origin or more pragmatic). But the modeling of real-world environmental (sound in landscape) behaviour touches on this, too. I know I am dangerously close to animism here – the idea of non-human organic and inorganic systems possessing 'will, choice and intention' may not be my intention to discuss here – but they certainly possess performative behaviour which we can 'play with' and respond to – and properly configured and addressed will respond to us.

My final world of 'presence' (the social) is paradoxically perhaps the one that does not seem to change much in the words of the retelling. 'What you mean to me, where I am and who I am with' will not change as a set of indicators of significance and meaning. But of course the 'you', the 'where' and the 'who' may be radically different in this model.

I may not know when performing if 'you' are human, computer, environmental, organic or inorganic – providing you are responsive! My sense of exactly where you are may also be no more than 'present for me, and responsive to me' – I have left open the third question 'who I am with' which unites these two - as one I do find intensely problematic with contemporary network technology. It is one which undoubtedly causes us some anxiety …

But anxiety is not a word to finish on. I find the possibilities – multiple and open – of a new musical world where technology connects up a wide variety of different sound makers in a responsive and animate network to be exciting and challenging - indeed I might have called my paper - 'Live Sounds in Organic Electroacoustic Music'.

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[12] The voice is a wonderful exception where we can create and control micro explosions, breakings, pops and clicks and yet *restore* the original materials – as Joan LaBarbara has said 'voice is the original instrument' with Trevor Wishart's paraphrase 'voice is the original synthesiser' [16].

[13] Or at least mental imagery – see http://www.emotiv.com/ - and their recent demo videos.

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